Quiz day 3

1. Using Gauss summation trick or otherwise find the sum

$$-5\binom{n}{0} - 3\binom{n}{1} - 1 \cdot \binom{n}{2} + 1 \cdot \binom{n}{3} + \cdots + (2n-5)\binom{n}{n}.$$

2. Consider the equation $a + b + c = 10$ with integers $a \geq 2$, $b \geq 1$, $c \geq 3$.

   (a) How many solutions are there?

   (b) Which solution would you encode as $\star \mid \mid \star \star \star$?